Algorithm:

Step 1: Initialization

a. Input n (size of board)
b. Create:

   board[n][n] filled with '.'

   col[n] → tracks used columns

   diag1[2n-1] → tracks main diagonals (row - col + n - 1)

   diag2[2n-1] → tracks anti-diagonals (row + col)

Step 2: Start Recursion

a. Call: solve(0)

Step 3: Recursive Function: solve(row)

a. Base Case:
   if row == n:
   count++
   printBoard()
   return
   All queens placed successfully → valid solution
b. Try placing queen in each column:
   for c = 0 to n-1:

Step 4.: Safety Check (O(1))

Check if placing queen is valid:

if col[c] == false AND diag1[row - c + n - 1] == false AND diag2[row + c] == false

Step 5: Place Queen

a. board[row][c] = 'Q'
b. col[c] = true
c. diag1[row - c + n - 1] = true
d. diag2[row + c] = true

Step 6: Recurse for Next Row solve(row + 1)

Step 7: Backtrack

a. Undo placement:
   board[row][c] = '.'
   col[c] = false
   diag1[row - c + n - 1] = false
   diag2[row + c] = false

Step 8: Continue Trying All Possibilities